

An update on the dietary ligands of the AhR.

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BACKGROUND: Halogenated aromatic hydrocarbons including dioxins and non-halogenated polycyclic aromatic hydrocarbons are ligands of an aryl hydrocarbon receptor (AhR) and stimulate its transformation. Exposure to these environmental contaminants occurs mainly through diet. Recent articles demonstrated that certain food factors regulate the AhR transformation and expression of downstream drug-metabolizing enzymes. OBJECTIVE: To explain the actions of these food factors on the AhR transformation, as the mechanisms underlying are not fully understood. METHODS: This review introduces recent articles that have demonstrated the molecular mechanisms by which food factors regulate the AhR transformation and downstream drug-metabolizing enzymes. RESULTS/CONCLUSION: The role of classical ligands including dioxins as agonists of the receptor is well documented. As to the food factors, they act as antagonists because they basically suppress the AhR transformation by different mechanisms. Moreover, the fate and metabolism of food factors are important to understand their mechanisms.